TOP 50 LINUX COMMANDS FOR DEVOPS • READABLE CHEAT SHEET

Beginner-friendly reference for everyday Linux work, deployments, and troubleshooting.

How to use this guide

You do not need to memorize hundreds of commands to start DevOps. Learn what each command does, practice it safely, and return to this guide when you need it.

Safety reminder: Commands such as `rm -rf`, recursive `chown`, and service restarts can cause damage or downtime. Check the path, user, and target before executing them.

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1. Files & Directories

1. ls — List files and folders

Inspect application folders, hidden files, permissions, configs, and logs.

```
ls
ls -l
ls -la
ls -lh
ls -lah /var/log
```

2. cd — Change directory

Navigate between application, configuration, deployment, and log directories.

```
cd /var/log
cd ..
cd ~
cd -
cd /etc/nginx
```

3. pwd — Print working directory

Confirm your current location before running scripts or path-dependent commands.

```
pwd
# Example: /home/devops/project
```

4. mkdir — Create directories

Create deployment, configuration, backup, and log directories.

```
mkdir logs
mkdir -p app/config
mkdir -p /tmp/myapp/logs
```

5. touch — Create a file or update its timestamp

Create empty files, scripts, test files, or marker files.

```
touch file.txt
touch deployment.log
```

6. cp — Copy files or directories

Back up configuration files before making changes.

```
cp source.txt backup.txt
cp -r app/ app-backup/
cp nginx.conf nginx.conf.backup
```

7. mv — Move or rename files

Rename configs, move artifacts, and organize deployment files.

```
mv old.txt new.txt
mv app.log /tmp/
mv config.yaml config.yaml.backup
```

8. rm — Remove files or directories

Remove old artifacts, temporary files, caches, and stale deployments.

```
rm file.txt
rm -r folder/
rm -rf folder/
```

WARNING: Be extremely careful with rm -rf, especially as root. Verify the path first.

9. find — Find files by name, type, or size

Locate configs, logs, certificates, artifacts, and unexpectedly large files.

```
find /path -name "filename"
find /var/log -name "*.log"
find /var/log -type f -size +500M
```

10. du — Check file or directory size

Find which directory is consuming disk space.

```
du -sh folder/
du -sh /var/log/*
```

2. Reading & Searching Files

11. cat — Print a file's contents

Quickly inspect small configuration and system files.

```
cat file.txt
cat /etc/os-release
```

12. less — Read large files with scrolling

Prefer less over cat for very large logs.

```
less app.log
# Inside less: /search finds text; q exits
```

13. head — Show the first lines

Inspect the beginning of logs, CSV files, and generated output.

```
head app.log
head -n 20 app.log
head -n 50 config.txt
```

14. tail — Show the last lines

Inspect recent application or system events.

```
tail app.log
tail -n 100 app.log
tail -n 50 /var/log/syslog
```

15. tail -f — Follow a log as it grows

Watch new log entries while testing an issue or deployment.

```
tail -f app.log
tail -f /var/log/nginx/access.log
```

16. grep — Search text for a pattern

Find errors, request details, IPs, and configuration references.

```
grep "ERROR" app.log
grep -i "error" app.log
grep -R "database_url" .
```

17. wc — Count lines, words, and bytes

Count errors, requests, entries, or generated records.

```
wc file.txt
wc -l file.txt
grep 'ERROR' app.log | wc -l
```

18. sort — Sort lines of text

Sort command output or log data before analyzing it.

```
sort names.txt
sort -n numbers.txt
sort access.log
```

19. uniq — Remove or count adjacent duplicates

Find repeated values such as IP addresses. Usually combine with sort.

```
sort file.txt | uniq
sort file.txt | uniq -c
cut -d' ' -f1 access.log | sort | uniq -c
```

20. awk — Extract or process text columns

Extract fields such as IP addresses, status codes, or columns from output.

```
awk '{print $1}' file.txt
awk '{print $1}' access.log
```

21. sed — Transform text or replace strings

Automate text changes in scripts and pipelines.

```
sed 's/old/new/g' file.txt
sed 's/dev/prod/g' config.txt
sed -i 's/dev/prod/g' config.txt
```

Check the target file before using sed -i because it edits the file in place.

3. Processes & System Health

22. ps — Show running processes

Check whether an application process is running.

```
ps
ps aux
ps aux | grep nginx
```

23. top — Monitor CPU and memory use

Investigate processes consuming resources when a server is slow. Press q to exit.

```
top
```

24. free — Check memory

Check whether the server is low on RAM. The available-memory figure is often more useful than free memory alone.

```
free -h
```

25. df — Check filesystem disk space

Investigate full disks that can prevent apps from writing logs or files.

```
df -h
df -h /
```

26. uptime — Show uptime and load averages

Quickly check how long the server has been running and its load averages.

```
uptime
```

27. kill — Send a signal to a process

Stop a stuck process. Try a normal kill first; use -9 only if necessary.

```
kill PID
kill -9 PID
```

28. lsof — List open files and network sockets

Find which process is using a file or port, especially for “Address already in use.”

```
lsof
sudo lsof -i :8080
```

4. Networking & Debugging

29. ip — Inspect interfaces and routes

Inspect IP addresses, interfaces, and routes.

```
ip addr
ip route
ip addr show
```

30. ping — Test basic network reachability

Test connectivity. A failed ping does not prove a service is down because ICMP may be blocked.

```
ping google.com
ping -c 4 google.com
```

31. curl — Send HTTP requests and debug APIs

Test APIs, health checks, proxies, and web services.

```
curl https://example.com
curl -I https://example.com
curl -v https://example.com
curl http://localhost:8080/health
```

32. wget — Download a file

Download binaries, artifacts, packages, or test files from a URL.

```
wget https://example.com/app.tar.gz
```

33. ss — Inspect listening ports and connections

Check whether an app is listening on the expected port. Process details may require sudo.

```
ss -tuln
sudo ss -ltnp | grep 8080
```

34. dig — Troubleshoot DNS

Inspect DNS answers and troubleshoot name resolution.

```
dig example.com
dig +short example.com
```

35. nslookup — Look up DNS records

Quickly check the IP address returned for a hostname.

```
nslookup example.com
```

36. traceroute — Trace the network path

Inspect network hops toward a destination. It may need installation and some hops may not respond.

```
traceroute example.com
```

37. hostname — Show server identity

Confirm which server you are connected to before making changes.

```
hostname
hostname -I
```

5. Users & Permissions

38. whoami — Show the current user

Verify the account you are using before privileged commands.

```
whoami
```

39. sudo — Run a command with elevated privileges

Perform administrative tasks without logging in directly as root. Use only when needed.

```
sudo systemctl restart nginx
```

40. chmod — Change file permissions

Make scripts executable and correct permission issues. Apply the least permissions required.

```
chmod 755 script.sh
chmod +x deploy.sh
```

41. chown — Change file owner and group

Correct ownership issues. Verify the path before using recursive -R.

```
sudo chown user:group file.txt
sudo chown nginx:nginx /var/www/app
sudo chown -R nginx:nginx /var/www/app
```

42. id — Show user and group details

Debug access problems and group membership.

```
id
id username
```

6. Services & Logs

43. systemctl — Manage systemd services

Check, start, stop, restart, or enable services on systemd-based Linux systems.

```
systemctl status nginx
sudo systemctl start nginx
sudo systemctl stop nginx
sudo systemctl restart nginx
sudo systemctl enable nginx
```

44. journalctl — Read systemd logs

Inspect service logs, follow them live, and narrow logs to a time window.

```
journalctl
journalctl -u nginx
journalctl -u nginx -f
journalctl -u nginx --since '10 minutes ago'
```

7. Archives & Transfers

45. tar — Create and extract archives

Package application files, logs, configs, and backups.

```
tar -czf backup.tar.gz app/
tar -xzf backup.tar.gz
```

46. scp — Copy files over SSH

Transfer files securely between local and remote machines.

```
scp file.txt user@server:/tmp/
scp user@server:/var/log/app.log .
```

47. rsync — Synchronize files efficiently

Sync deployment files and backups. Review destination paths before syncing.

```
rsync -av source/ destination/
rsync -av ./app/ user@server:/var/www/app/
```

8. Remote Servers

48. ssh — Connect to a remote server

Connect to Linux servers for administration, debugging, and incident response. Keep private keys secure.

```
ssh user@server
ssh ubuntu@10.0.1.20
ssh -i key.pem ubuntu@server
```

9. Command-Line Superpowers

49. history — View previous commands

Find commands you ran earlier instead of retyping them.

```
history
history | grep kubectl
# Ctrl + R searches interactively
```

50. man — Open a command manual

Learn options when you forget a flag; you do not need to memorize every option.

```
man grep
grep --help
```

Bonus: Useful DevOps Command Combinations

Find errors in a log

```
grep -i "error" app.log
```

Watch errors live

```
tail -f app.log | grep -i "error"
```

Find what is using port 8080

```
sudo lsof -i :8080
# Alternative:
sudo ss -ltnp | grep 8080
```

Find the largest items in the current directory

```
du -sh -- * | sort -h
```

Find huge log files

```
find /var/log -type f -size +500M
```

Check Nginx status

```
systemctl status nginx
```

Follow Nginx service logs

```
journalctl -u nginx -f
```

Check whether an API responds

```
curl -I https://example.com
```

Debug an API request

```
curl -v https://example.com/api/health
```

Find an Nginx process

```
ps aux | grep nginx
```

Count errors in a log

```
grep -i "error" app.log | wc -l
```

Check CPU / memory / disk

```
top
free -h
df -h
```

Check DNS and listening ports

```
dig example.com
ss -tuln
```

The DevOps Troubleshooting Mindset

Start by identifying the symptom, then run a small set of targeted checks. Avoid changing production settings until you understand what the output means.

Problem	Useful first checks
Application is down	systemctl status app ps aux | grep app
Application is crashing	journalctl -u app --since '10 minutes ago' tail -f app.log
Server is slow	top free -h uptime
Disk is full	df -h du -sh -- /* (may need sudo)

Problem	Useful first checks
Port is not reachable	ss -ltnp curl -v http://localhost:8080
DNS is failing	dig example.com nslookup example.com
Cannot find a file	find / -name "filename" 2>/dev/null
Permission denied	ls -l whoami id

Quick Reference: All 50 Commands

FILES & DIRECTORIES

01 ls — list files; 02 cd — change directory; 03 pwd — current directory; 04 mkdir — create directory; 05 touch — create/update file; 06 cp — copy; 07 mv — move/rename; 08 rm — delete; 09 find — locate files; 10 du — disk usage.

READ & SEARCH

11 cat — print file; 12 less — read large files; 13 head — first lines; 14 tail — last lines; 15 tail -f — follow logs; 16 grep — search text; 17 wc — count; 18 sort — sort lines; 19 uniq — adjacent duplicates; 20 awk — process fields; 21 sed — transform text.

SYSTEM HEALTH

22 ps — processes; 23 top — resource monitor; 24 free — memory; 25 df — disk space; 26 uptime — uptime/load; 27 kill — signal process; 28 lsof — open files/sockets.

NETWORKING

29 ip — interfaces/routes; 30 ping — reachability; 31 curl — HTTP/API; 32 wget — download; 33 ss — sockets/ports; 34 dig — DNS; 35 nslookup — DNS; 36 traceroute — network path; 37 hostname — server identity.

PERMISSIONS

38 whoami — current user; 39 sudo — elevated command; 40 chmod — permissions; 41 chown — ownership; 42 id — user/groups.

SERVICES / TRANSFERS / HELP

43 systemctl — services; 44 journalctl — systemd logs; 45 tar — archive/extract; 46 scp — secure copy; 47 rsync — sync; 48 ssh — remote login; 49 history — command history; 50 man — manual.

Recommended learning order: ls → cd → pwd → cat → grep → tail -f → ps → top → df -h → systemctl → journalctl → ss → curl → dig → ssh

Tools get you started. Troubleshooting makes you a DevOps engineer.